

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

DECEMBER EXAMINATION

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES

MONDAY, 23 DECEMBER 2024

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATION AND NETWORKING

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **CHOOSE ANY ONE (1)** question in **Section B**.
Each question carries 25 marks each.

Note: Should you answer more than ONE (1) question in section B, marks will only be awarded to the **FIRST** question attempted.

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES**Answer ALL Questions:**

Q1. (a) Read the following case study:

Marcus is a software development team lead. He is assigned to develop a mobile App for a food delivery service. A customer from XYZ Restaurant Chain requires Marcus's team to complete the project quickly due to increasing market competition. Referring to this scenario, answer the following questions:

- (i) Suggest the most appropriate development approach. Provide **TWO (2)** justifications for the suggested solution. (8 marks)
- (ii) Discuss **TWO (2)** disadvantages of the approach used in Q1 (a (i)). (6 marks)

(b) Referring to the food delivery App project in Q1(a), the development timeline spans 10 months and 3 weeks, broken down into 8 sprint cycles. The development teams work 9 hours daily, collaborating across Asian and European time zones. According to the scenario, answer the following questions:

- (i) Calculate a period time required for each sprint. **Note: The duration for each sprint is measured in weeks and prove your calculations in Q1 (b)(i).** (3 marks)
- (ii) As a scrum master, do you agree with the estimated sprint period in (b)(i)? Justify your answer. (3 marks)
- (iii) If the development lead allocates 3 weeks for each sprint, does this mean all sprint cycles will strictly follow this 3-week duration throughout the project? Justify your answer. **Note: this question is not related to previous calculations in Q1(b)(i).** (3 marks)
- (iv) State **TWO (2)** main responsibilities of the development team during the daily stand-up meetings. (2 marks)

[Total : 25 marks]

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Q2. (a) Read the following case study:

A busy international airport has implemented a dynamic baggage handling system to manage thousands of pieces of luggage daily across multiple terminals.

- (i) Using smart tracking devices, ground staff can monitor baggage locations and movements status across the entire airport network.
- (ii) The system's advanced sorting algorithms automatically route baggage to correct destinations, handling transfer flights and priority baggage efficiently.
- (iii) At the control center, operators oversee operations through digital maps and status boards, managing exceptions and ensuring smooth baggage flow between check-in, security screening, and loading zones.

The baggage handling system has drastically reduced mishandled luggage incidents while improving operational efficiency, enabling the airport to process increasing passengers' volumes while maintaining reliable baggage delivery services.

According to the case study above, propose **ONE (1)** most appropriate architectural pattern, and justify your proposed answer based on its **TWO (2)** advantages. (5 marks)

(b) Read the following case study:

The TravelPass app is gaining popularity as a digital platform for booking and managing intercity bus travel. Passengers use the app to search routes, purchase tickets, and access travel information. The app features include seat selection, multiple payment methods, digital ticket storage, and live bus tracking.

Your development team has been assigned to enhance the TravelPass app with new capabilities. To ensure clear development objectives, you must create detailed user stories to guide the implementation process.

Provide **FIVE (5)** user stories for the TravelPass application. The user stories should address different functionalities including route searching, ticket booking, seat selection, and travel management. (10 marks)

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES**Q2. (Continued)**

- (c) The TravelPass App in Q2(b) needs to integrate with external components as follows:

PaymentGateway component:

Processes online payments
Validates transactions
Generates payment receipts

BusOperator component:

Provides real-time bus location
Updates seat availability
Manages schedule changes

Referring to this scenario, answer the following questions:

- (i) Identify **TWO (2)** potential interface incompatibilities between these components and explain how you would resolve them using appropriate adaptation techniques. (6 marks)
- (ii) Explain **TWO (2)** ways to ensure reliable data synchronization between these components when the network connection is unstable. (4 marks)
- [Total : 25 marks]

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Q3. (a) Read the following scenario:

Your team is developing components for a hospital information system. Two key components have been identified as follows:

PatientRecord component:

- Stores and manages patient medical histories.
- Handles appointment scheduling.
- Maintains prescription records.

BillingSystem component:

- Processes insurance claims.
- Generates patient invoices.
- Manages payment records.

Referring to this scenario, answer the following questions:

- Identify **TWO (2)** potential interface incompatibilities between these components and explain how you would resolve them using appropriate adaptation techniques. (6 marks)
- If the hospital later wants to replace the BillingSystem component with a third-party solution, explain **THREE (3)** component validation checks that should be performed before integration. (9 marks)

(b) The hospital plans to add a new PharmacyInventory component that needs to interact with both existing components. This component will function as follows:

- Track medication inventory levels.
- Process prescription requests.
- Generate reorder alerts.
- Record medication dispensing.

Referring to this scenario, answer the following questions:

- Design a composition diagram showing how the PharmacyInventory component will interact with the existing components. Include the interfaces and any necessary adapters. (5 marks)
- Explain **TWO (2)** principles of component reusability that should be considered when developing the PharmacyInventory component to ensure it can be reused in other healthcare systems. (5 marks)

[Total : 25 marks]

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES**SECTION B (Answer ONE Question)**

Q4. (a) WeatherWatch is a weather monitoring system that collects data from multiple weather stations:

- The system records temperature, humidity, and wind speed measurements from each station.
- It processes readings every hour and generates weather alerts.
- Weather forecasters use the system to view current conditions and create daily forecast reports.

Referring to this scenario, answer the following questions:

- (i) Based on the development testing covered in the lecture, identify and explain **THREE (3)** levels of testing that should be performed. Provide a specific example for each level relevant to the WeatherWatch system. (9 marks)
- (ii) Identify **TWO (2)** different types of interface errors discussed in the lecture that might occur in this system and explain how each could be tested. (6 marks)

- (b) A program module in WeatherWatch in Q4(a) accepts temperature readings with the following requirements:
- Must handle between 4 to 10 station inputs.
 - Temperature values must be between -50°C to +50°C.
 - Only integer values are accepted.

Referring to this scenario, answer the following questions:

- (i) Using the partition testing approach covered in the lecture, identify the input partitions and suggest **THREE (3)** appropriate test cases. Show your working. (5 marks)
- (ii) With reference to the lecture material, explain **TWO (2)** key differences between release testing and system testing in the context of this weather monitoring system. (5 marks)

[Total : 25 marks]

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- Q5. (a) Consider a university's Student Information System (SIS) that has been operating for 10 years. The development team needs to test new features being added to the registration module.

Referring to this scenario, answer the following questions:

- (i) Using the three levels of development testing discussed in the lecture, explain how you would test the new registration features. Provide specific examples for each level. (6 marks)
 - (ii) Using partition testing, identify the input partitions and suggest appropriate test cases for the following requirement: "The system must handle between 5 to 200 concurrent student registrations, where each student can register for 1 to 6 courses per semester." (4 marks)
- (b) The university has determined that the SIS in Q5(a) is a legacy system that needs attention. Referring to this statement, answer the following questions:
- (i) Identify **THREE (3)** legacy system components present in the current SIS and explain the challenges associated with each. (6 marks)
 - (ii) Based on the system quality and business value assessment covered in the lecture, recommend whether to scrap, maintain, reengineer or continue operating the system. Justify your answer. (4 marks)
- (c) You are appointed as the project manager to oversee the modernization of the Student Information System (SIS). Referring to this statement, answer the following questions:
- (i) Analyze **TWO (2)** potential risks from the following risk types: people, technology, and organizational by stating its probability (very low, low, moderate, high, very high) and its effect (catastrophic, serious, tolerable, insignificant). (4 marks)
 - (ii) Select **ONE (1)** risk identified in Q5(c)(i) and propose an appropriate risk management strategy. (1 mark)

[Total : 25 marks]

